CBC Digital Learning Platform

Stage B — Use Case Modelling

UML Use Case Diagrams & Descriptions

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Prepared by: Project Team — End of Year Project

Status: Draft — Design Phase

1. Introduction

This document presents the Stage B Use Case Modelling output for the CBC Digital Learning Platform. It derives from the Software Requirements Specification (SRS v1.0) and transforms all functional requirements (FR-L, FR-T, FR-M, FR-AI) into structured UML Use Case Diagrams and formal use case descriptions.

The use cases define every meaningful interaction between the system's actors and the platform, establishing the behavioural baseline for subsequent design stages (Domain Modelling, Architecture, Database Design, and UI/UX Design).

1.1 Actor Summary

Five actors have been identified and modelled across four use case diagrams:

Actor	Type	Description	Diagram
User (Generalised)	Primary — Abstract	Generalised actor owning shared authentication behaviour inherited by all human actors.	Diagram 1
Learner	Primary — Human	Secondary school student (S1–S6). Central user of the platform.	Diagram 2
Teacher	Primary — Human	Subject teacher in a registered secondary school.	Diagram 3
MoES/NCDC Official	Primary — Human	Ministry and curriculum authority staff with governance responsibility.	Diagram 4
AI Tutor	Secondary — System	Embedded AI assistant responding to learner academic queries via MCP knowledge base.	Diagram 4

1.2 UML Conventions Applied

The following standard UML conventions have been applied throughout this document:

Convention	Application
Actor Generalisation	«User» generalised actor owns Register, Log In, Log Out. Learner and Teacher inherit via hollow-triangle arrows. MoES/NCDC inherits Log In and Log Out only (no self-registration).
«include»	Used where a use case unconditionally requires the execution of another (e.g., Access Lesson Offline always includes Download Lesson).
«extend»	Used where a use case conditionally extends another at an extension point (e.g., Edit Posted Material optionally extends View Posted Materials).

Convention	Application
System Boundary	Dashed rectangle enclosing all use cases for each subsystem. Actors are placed outside the boundary.
Secondary System Actor	AI Tutor rendered as a labelled rectangle (not stick figure) on the right side of the system boundary, per UML convention for non-human system actors.

2. Use Case Diagrams

Diagram 1 — Authentication (Generalised User Actor)

This diagram captures the shared authentication use cases owned by the generalised «User» actor. The Learner and Teacher actors inherit these behaviours via actor generalisation. MoES/NCDC officials are onboarded administratively and therefore do not use Register Account.

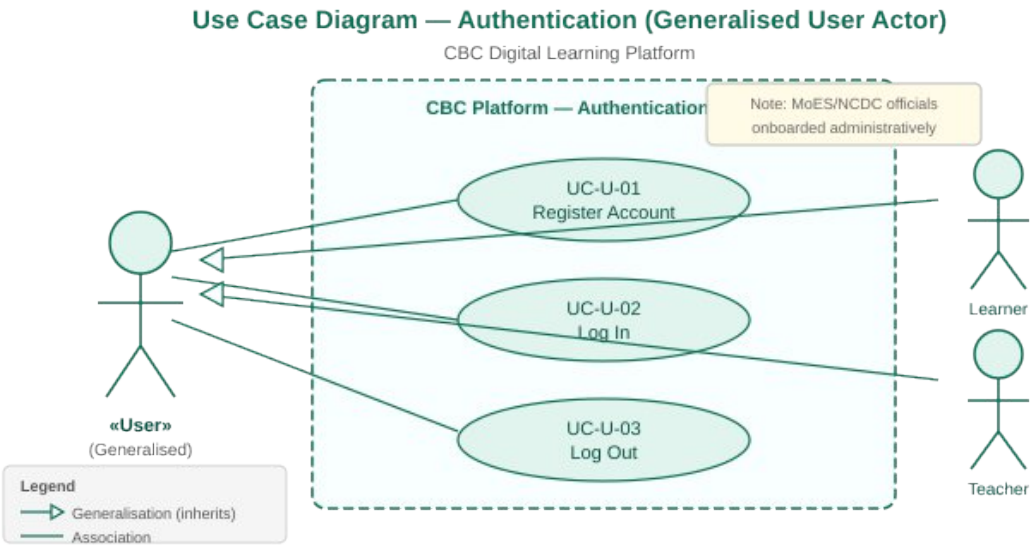


Diagram 2 — Learner

This diagram captures all use cases available to the Learner role, derived from FR-L-03 through FR-L-11. The Learner inherits authentication behaviour from the generalised «User» actor (shown in Diagram 1). An «include» dependency between Access Lesson Offline and Download Lesson for Offline Access reflects the mandatory prerequisite relationship between these two use cases.

Use Case Diagram — Learner

CBC Digital Learning Platform

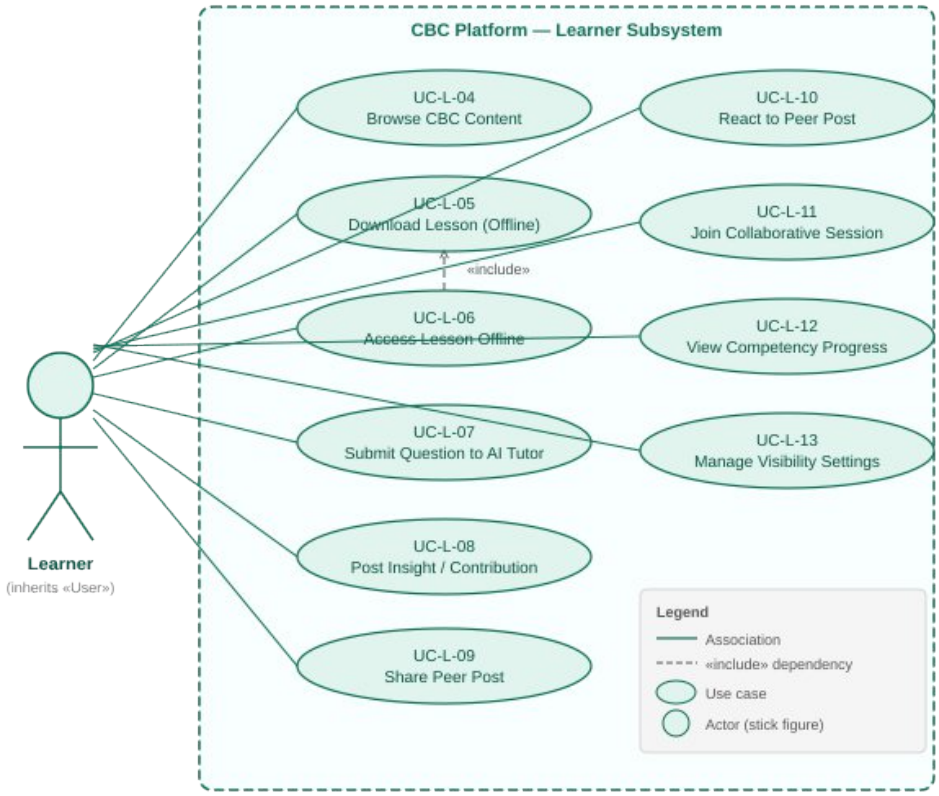


Diagram 3 — Teacher

This diagram captures all use cases available to the Teacher role, derived from FR-T-02 through FR-T-04, plus analyst additions for material management. The Teacher inherits authentication behaviour from «User». «include» relationships model the mandatory tagging and sharing steps within the posting flow. «extend» relationships model the optional edit and remove actions triggered from viewing posted materials.

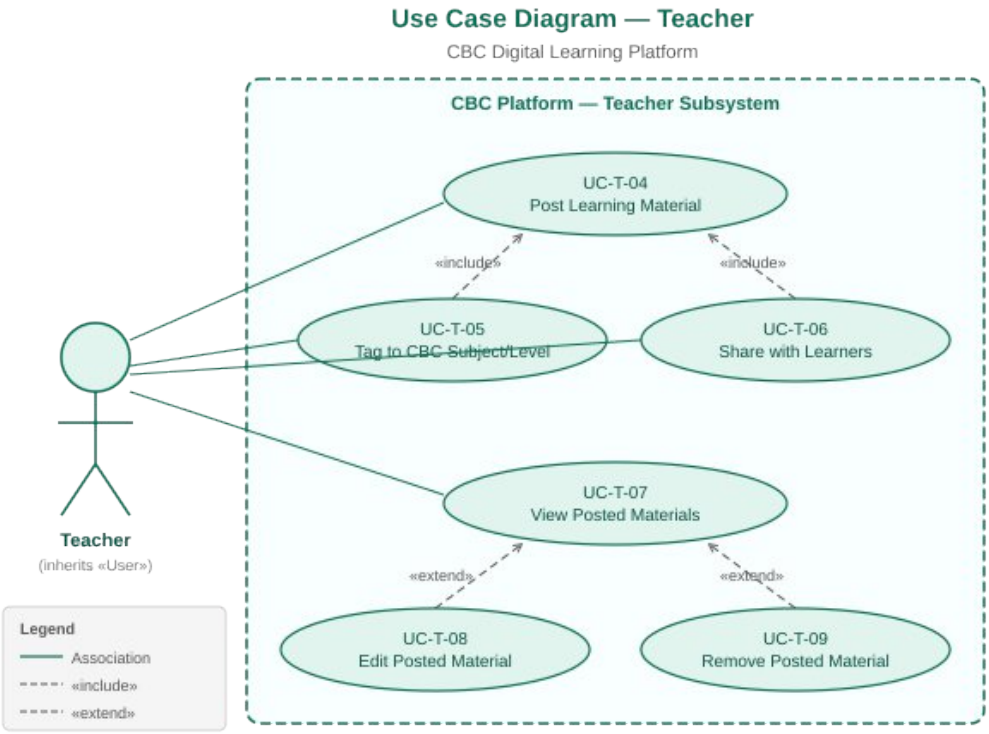
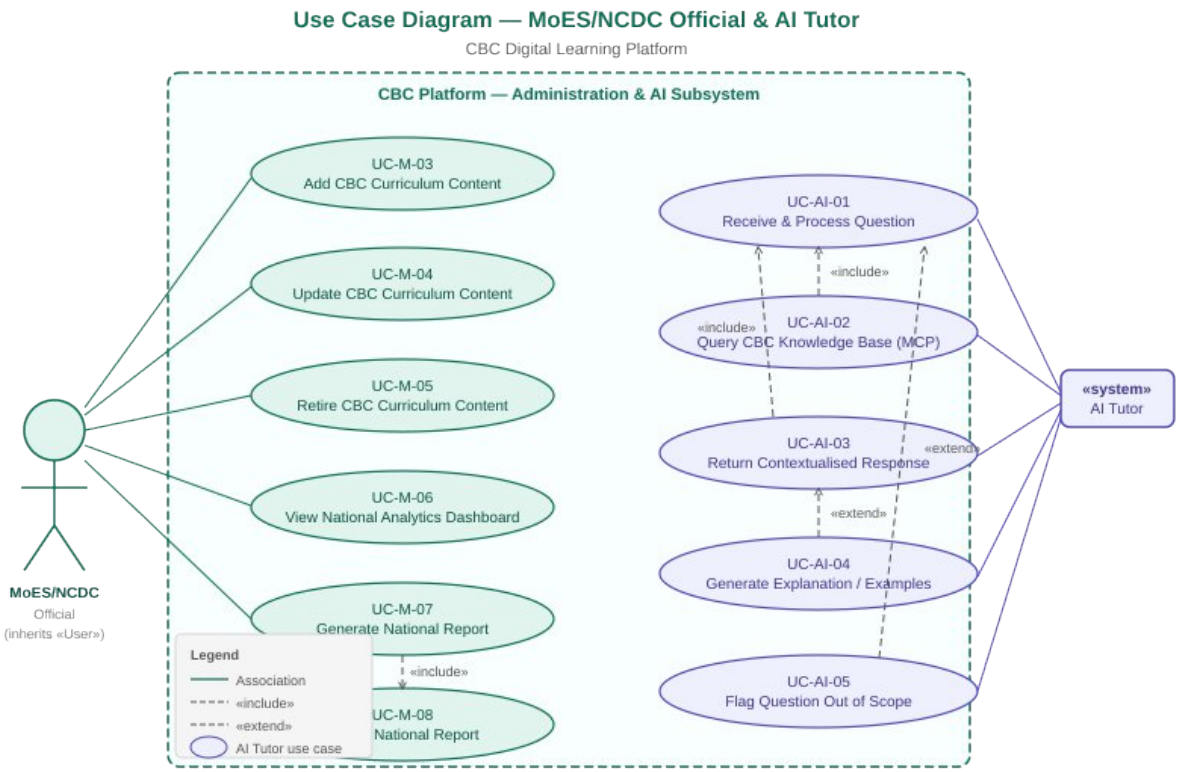


Diagram 4 — MoES/NCDC Official & AI Tutor

This diagram combines the MoES/NCDC Official and AI Tutor actors on a single system boundary. The MoES/NCDC Official manages curriculum content and national reporting. The AI Tutor is modelled as a secondary system actor (rectangle) on the right of the boundary, per UML convention. The «include» and «extend» relationships within the AI Tutor use cases reflect mandatory knowledge base querying and conditional out-of-scope flagging behaviour.



3. Use Case Descriptions

The following tables provide structured descriptions for all use cases. Each description captures the actor, preconditions, main flow, alternate/exception flows, and postconditions — forming the behavioural specification for each interaction.

3.1 Authentication Use Cases

UC-U-01 Register Account

Field	Description
Use Case ID	UC-U-01
Use Case Name	Register Account
Actor	Learner, Teacher
Preconditions	User is not yet registered. School has been onboarded by an administrator.
Main Flow	1. User navigates to the registration page. 2. User enters username and password. 3. System validates inputs (unique username, password strength). 4. System creates account and assigns role. 5. System confirms successful registration.
Alternate Flow	A1: Username already exists — system displays error, prompts user to choose a different username.
Exception Flow	E1: System unavailable — registration fails; user is informed to retry.
Postconditions	User account exists in the system with assigned role (Learner or Teacher).
Priority	High — Phase 1
Derived From	FR-L-01, FR-T-01

UC-U-02 Log In

Field	Description
Use Case ID	UC-U-02
Use Case Name	Log In
Actor	Learner, Teacher, MoES/NCDC Official
Preconditions	User has a registered account. For MoES/NCDC: account provisioned administratively.
Main Flow	1. User navigates to the login page. 2. User enters username and password. 3. System authenticates credentials against hashed password store. 4. System starts an authenticated session and redirects to role-appropriate dashboard.
Alternate Flow	A1: Invalid credentials — system displays error. User may retry.

Field	Description
Exception Flow	E1: System unavailable — login fails; user is informed.
Postconditions	Authenticated session established. User is on their role-specific dashboard.
Priority	High — Phase 1
Derived From	FR-L-02, FR-T-01, FR-M-01

UC-U-03 Log Out

Field	Description
Use Case ID	UC-U-03
Use Case Name	Log Out
Actor	Learner, Teacher, MoES/NCDC Official
Preconditions	User has an active authenticated session.
Main Flow	1. User selects Log Out. 2. System terminates the session and clears session tokens. 3. System redirects user to the login page.
Alternate Flow	None.
Exception Flow	E1: Session already expired — system redirects silently to login page.
Postconditions	Session is terminated. No authenticated state persists.
Priority	High — Phase 1
Derived From	FR-L-02, FR-T-01 (Convention — industry standard)

3.2 Learner Use Cases

UC-L-04 Browse CBC Content by Subject & Level

Field	Description
Use Case ID	UC-L-04
Use Case Name	Browse CBC Content by Subject & Level
Actor	Learner
Preconditions	Learner is logged in.
Main Flow	1. Learner selects a subject from the curriculum menu. 2. Learner selects their class level (S1–S6). 3. System displays curriculum content organised by competency. 4. Learner selects a lesson to open.
Alternate Flow	A1: No content available for selected subject/level — system displays a notice.
Exception Flow	E1: No connectivity — system offers access to previously downloaded lessons only (see UC-L-06).
Postconditions	Learner is viewing selected lesson content.
Priority	High — Phase 1
Derived From	FR-L-03

UC-L-05 Download Lesson for Offline Access

Field	Description
Use Case ID	UC-L-05
Use Case Name	Download Lesson for Offline Access
Actor	Learner
Preconditions	Learner is logged in. Active internet connection is available.
Main Flow	1. Learner opens a lesson (via UC-L-04). 2. Learner selects Download for Offline Access. 3. System compresses and caches the lesson content to local storage. 4. System confirms the lesson is available offline.
Alternate Flow	A1: Insufficient local storage — system warns learner and aborts download.
Exception Flow	E1: Connection lost mid-download — download fails; learner is notified to retry.
Postconditions	Lesson is stored locally and marked as available offline.
Priority	High — Phase 1-2
Derived From	FR-L-04

UC-L-06 Access Lesson Offline

Field	Description
Use Case ID	UC-L-06
Use Case Name	Access Lesson Offline
Actor	Learner
Preconditions	Learner has previously downloaded the lesson (UC-L-05). No active internet connection required.
Includes	UC-L-05 (Download Lesson for Offline Access) — must have executed prior to this use case.
Main Flow	1. Learner opens the platform while offline. 2. System detects offline state and displays offline indicator. 3. Learner navigates to Downloaded Lessons. 4. Learner opens a previously downloaded lesson. 5. System renders lesson from local cache.
Alternate Flow	A1: No lessons downloaded — system informs learner; online features unavailable.
Exception Flow	None specific beyond offline indicator display.
Postconditions	Learner is reading lesson content from local cache without internet connectivity.
Priority	High — Phase 1-2
Derived From	FR-L-04, NFR-O-01

UC-L-07 Submit Question to AI Tutor

Field	Description
Use Case ID	UC-L-07
Use Case Name	Submit Question to AI Tutor
Actor	Learner (initiator), AI Tutor (respondent)
Preconditions	Learner is logged in. Active internet connection is available.
Main Flow	1. Learner types an academic question in the AI Tutor input panel. 2. Learner submits the question. 3. System forwards the query to the AI Tutor engine. 4. AI Tutor queries the CBC Knowledge Base via MCP (UC-AI-02). 5. AI Tutor returns a contextualised response within 5 seconds (NFR-P-02). 6. System displays the response inline in the learner's session.
Alternate Flow	A1: Question falls outside the knowledge base — AI Tutor flags this (UC-AI-05) and advises the learner accordingly.
Exception Flow	E1: AI layer unavailable — system informs learner; core content access remains functional (NFR-A-03).
Postconditions	Learner has received a response. Interaction is logged (private by default, per NFR-SEC-05).

Field	Description
Priority	High — Phase 4
Derived From	FR-L-05, FR-AI-01 to FR-AI-05

UC-L-08 Post Insight / Research Contribution

Field	Description
Use Case ID	UC-L-08
Use Case Name	Post Insight / Research Contribution
Actor	Learner
Preconditions	Learner is logged in. Active internet connection is available.
Main Flow	1. Learner navigates to the Knowledge Sharing section. 2. Learner composes a post (insight, idea, or research contribution). 3. Learner sets visibility (private by default — see UC-L-13). 4. Learner submits the post. 5. System saves and publishes the post per the visibility setting.
Alternate Flow	A1: Post is empty — system prompts learner to add content before submitting.
Exception Flow	E1: Connection lost before submission — draft is preserved locally; learner is notified.
Postconditions	Post is saved and visible according to the learner's visibility settings.
Priority	High — Phase 3
Derived From	FR-L-06

UC-L-09 / UC-L-10 Share & React to Peer Post

Field	Description
Use Case ID	UC-L-09, UC-L-10
Use Case Name	Share Peer Post / React to Peer Post
Actor	Learner
Preconditions	Learner is logged in. Peer post is visible to the learner per its visibility setting.
Main Flow (UC-L-09)	1. Learner views a peer's post in the knowledge feed. 2. Learner selects Share. 3. System shares the post to the learner's own profile or to selected peers. 4. System records the share action.
Main Flow (UC-L-10)	1. Learner views a peer's post. 2. Learner selects a reaction (e.g., helpful, insightful). 3. System records the reaction and updates the post's engagement count.
Exception Flow	E1: Connection lost — action is queued and retried when connectivity

Field	Description
	is restored.
Postconditions	Share or reaction is recorded and reflected in the platform's engagement data.
Priority	High — Phase 3
Derived From	FR-L-07

UC-L-11 Join Collaborative Learning Session

Field	Description
Use Case ID	UC-L-11
Use Case Name	Join Collaborative Learning Session
Actor	Learner
Preconditions	Learner is logged in. Active internet connection. A session is available.
Main Flow	1. Learner navigates to the Collaboration section. 2. Learner browses or searches for an active session. 3. Learner selects a session and joins. 4. System adds learner to the real-time session. 5. Learner participates in shared content, discussion, or peer interaction.
Alternate Flow	A1: Session is full or restricted — system informs learner.
Exception Flow	E1: Connection drops during session — system attempts reconnection and resumes session state.
Postconditions	Learner has participated in a collaborative learning session.
Priority	High — Phase 3
Derived From	FR-L-08

UC-L-12 View Competency Progress Dashboard

Field	Description
Use Case ID	UC-L-12
Use Case Name	View Competency Progress Dashboard
Actor	Learner
Preconditions	Learner is logged in. At least some platform activity has been recorded.
Main Flow	1. Learner navigates to their Progress Dashboard. 2. System retrieves and displays competency attainment data aligned to the CBC framework. 3. Learner views progress per subject, level, and competency.
Alternate Flow	A1: No activity recorded yet — system displays an empty state with guidance.

Field	Description
Exception Flow	None.
Postconditions	Learner has viewed their current CBC competency attainment status.
Priority	High — Phase 2
Derived From	FR-L-09

UC-L-13 Manage Post Visibility Settings

Field	Description
Use Case ID	UC-L-13
Use Case Name	Manage Post Visibility Settings
Actor	Learner
Preconditions	Learner is logged in. Learner has at least one post.
Main Flow	1. Learner navigates to a post or the visibility settings panel. 2. Learner selects the visibility level: Private (default), Peers, Teachers, or Public. 3. System saves the visibility setting. 4. System enforces the setting on all content access attempts.
Alternate Flow	None.
Exception Flow	None.
Postconditions	Post visibility is updated and enforced by the access control layer.
Priority	High — Phase 1
Derived From	FR-L-10, FR-L-11, NFR-SEC-05

3.3 Teacher Use Cases

UC-T-04 Post Learning Material

Field	Description
Use Case ID	UC-T-04
Use Case Name	Post Learning Material
Actor	Teacher
Preconditions	Teacher is logged in.
Main Flow	1. Teacher navigates to the Content Management section. 2. Teacher creates a new material post (lesson, guide, resource). 3. Teacher tags the material to CBC subject, level and competency (UC-T-05 — included). 4. Teacher shares the material with registered learners (UC-T-06 — included). 5. System saves and publishes the material.
Alternate Flow	A1: Required fields incomplete — system prompts teacher to complete tagging before posting.
Exception Flow	E1: Upload fails — system retains draft; teacher is notified.
Postconditions	Material is published and visible to the designated learners.
Priority	High — Phase 2
Derived From	FR-T-02

UC-T-07 / UC-T-08 / UC-T-09 View, Edit & Remove Posted Material

Field	Description
Use Case ID	UC-T-07, UC-T-08, UC-T-09
Use Case Name	View Posted Materials / Edit Posted Material / Remove Posted Material
Actor	Teacher
Preconditions	Teacher is logged in. At least one material has been posted.
Main Flow (UC-T-07)	1. Teacher navigates to My Materials. 2. System displays all materials posted by the teacher. 3. Teacher reviews the list. (Optionally triggers UC-T-08 or UC-T-09.)
Main Flow (UC-T-08 — extends UC-T-07)	1. Teacher selects Edit on a material. 2. Teacher modifies content, tagging, or sharing settings. 3. System saves the updated material.
Main Flow (UC-T-09 — extends UC-T-07)	1. Teacher selects Remove on a material. 2. System requests confirmation. 3. Teacher confirms. 4. System removes the material and revokes learner access.
Exception Flow	E1: Material has been shared — system warns teacher that removing it will revoke learner access.
Postconditions	UC-T-07: Teacher has viewed materials. UC-T-08: Material is updated. UC-T-09: Material is removed from the platform.

Field	Description
Priority	Medium — Analyst addition, Phase 2
Derived From	Analyst addition (industry standard CRUD convention)

3.4 MoES/NCDC Official Use Cases

UC-M-03 to UC-M-05 Curriculum Content Management

Field	Description
Use Case ID	UC-M-03, UC-M-04, UC-M-05
Use Case Name	Add / Update / Retire CBC Curriculum Content
Actor	MoES/NCDC Official
Preconditions	Official is logged in with MoES/NCDC role-based access.
Main Flow (UC-M-03 Add)	1. Official navigates to Curriculum Management. 2. Official creates a new content record (subject, level, competency, material). 3. System validates and ingests content into the CBC Knowledge Base. 4. System confirms successful addition.
Main Flow (UC-M-04 Update)	1. Official selects existing content. 2. Official edits content fields. 3. System saves changes and propagates update to the Knowledge Base.
Main Flow (UC-M-05 Retire)	1. Official selects content to retire. 2. System requests confirmation. 3. Official confirms. 4. System marks content as retired; it is removed from learner-facing views but retained in audit logs.
Exception Flow	E1: Content referenced by active learner sessions — system warns official before retirement.
Postconditions	CBC Knowledge Base reflects the official's changes. Learner-facing content is updated accordingly.
Priority	High — Phase 1-2
Derived From	FR-M-02

UC-M-06 to UC-M-08 Analytics & Reporting

Field	Description
Use Case ID	UC-M-06, UC-M-07, UC-M-08
Use Case Name	View Analytics Dashboard / Generate / Export National Report
Actor	MoES/NCDC Official
Preconditions	Official is logged in with MoES/NCDC role.
Main Flow (UC-M-06)	1. Official navigates to the National Analytics Dashboard. 2. System retrieves and displays platform usage, competency attainment trends, and engagement data by region and school.
Main Flow (UC-M-07 + UC-M-08)	1. Official selects Generate Report with desired filters (date range, region, subject). 2. System compiles report data. 3. System presents report preview. 4. Official selects Export (UC-M-08 — included). 5. System generates and downloads the report in the selected format.
Alternate Flow	A1: No data for selected filters — system informs official and suggests broadening the filter range.

Field	Description
Exception Flow	E1: Report generation times out for very large datasets — system queues the report and notifies the official when ready.
Postconditions	UC-M-06: Official has viewed national learning analytics. UC-M-07/08: Report is generated and exported successfully.
Priority	High (UC-M-06) / Medium (UC-M-07/08) — Phase 3-4
Derived From	FR-M-03, FR-M-04

3.5 AI Tutor Use Cases

UC-AI-01 to UC-AI-05 AI Tutor Behaviour

Field	Description
Use Case ID	UC-AI-01 to UC-AI-05
Use Case Name	AI Tutor: Receive, Process, Query, Respond, Flag
Actor	AI Tutor (system actor)
Preconditions	Learner has submitted a question (UC-L-07). AI layer is operational. MCP Knowledge Base is accessible.
Main Flow	1. AI Tutor receives the learner's question (UC-AI-01). 2. AI Tutor queries the CBC Knowledge Base via MCP server (UC-AI-02 — included). 3. AI Tutor constructs a contextualised response grounded in Ugandan CBC content (UC-AI-03 — included). 4. Where appropriate, AI Tutor generates concept explanations and local examples (UC-AI-04 — extends UC-AI-03). 5. Response is returned to the platform for display to the learner.
Alternate Flow (UC-AI-05)	A1: Knowledge base query returns no valid result — AI Tutor flags the question as outside the knowledge base (UC-AI-05 — extends UC-AI-01) and returns a transparent notice to the learner rather than generating a speculative response.
Exception Flow	E1: MCP server unavailable — AI Tutor falls back to a cached subset of knowledge if available, or informs the learner that the tutor is temporarily unavailable. Core content access is unaffected (NFR-A-03).
Postconditions	Learner receives either a grounded contextualised response or a transparent out-of-scope notice. Interaction is logged privately.
Priority	High — Phase 4
Derived From	FR-AI-01 to FR-AI-05

4. Requirements Traceability — Stage B

The table below confirms that every use case in Stage B traces back to at least one functional requirement in the SRS, and that every functional requirement is covered by at least one use case.

Use Case ID	Use Case Name	FR Requirement(s)	Priority
UC-U-01	Register Account	FR-L-01, FR-T-01	High
UC-U-02	Log In	FR-L-02, FR-T-01, FR-M-01	High
UC-U-03	Log Out	FR-L-02 (Convention)	High
UC-L-04	Browse CBC Content	FR-L-03	High
UC-L-05	Download Lesson for Offline Access	FR-L-04	High
UC-L-06	Access Lesson Offline	FR-L-04, NFR-O-01	High
UC-L-07	Submit Question to AI Tutor	FR-L-05, FR-AI-01–05	High
UC-L-08	Post Insight / Contribution	FR-L-06	High
UC-L-09	Share Peer Post	FR-L-07	High
UC-L-10	React to Peer Post	FR-L-07	High
UC-L-11	Join Collaborative Session	FR-L-08	High
UC-L-12	View Competency Progress	FR-L-09	High
UC-L-13	Manage Visibility Settings	FR-L-10, FR-L-11, NFR-SEC-05	High
UC-T-04	Post Learning Material	FR-T-02	High
UC-T-05	Tag Material to CBC Subject/Level	FR-T-03	High
UC-T-06	Share Material with Learners	FR-T-04	High
UC-T-07	View Posted Materials	Analyst addition	Medium
UC-T-08	Edit Posted Material	Analyst addition	Medium
UC-T-09	Remove Posted Material	Analyst addition	Medium
UC-M-03	Add CBC Curriculum Content	FR-M-02	High
UC-M-04	Update CBC Curriculum Content	FR-M-02	High
UC-M-05	Retire CBC Curriculum Content	FR-M-02	High
UC-M-06	View National Analytics Dashboard	FR-M-03	High

Use Case ID	Use Case Name	FR Requirement(s)	Priority
UC-M-07	Generate National Report	FR-M-04	Medium
UC-M-08	Export National Report	FR-M-04	Medium
UC-AI-01	Receive & Process Question	FR-AI-01	High
UC-AI-02	Query CBC Knowledge Base (MCP)	FR-AI-03	High
UC-AI-03	Return Contextualised Response	FR-AI-02, FR-AI-04	High
UC-AI-04	Generate Explanation / Examples	FR-AI-04	High
UC-AI-05	Flag Question Out of Scope	FR-AI-05	High